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Displays with silicon and semiconducting oxide thin-film transistors

Abstract

An electronic device may include a display having an array of display pixels on a substrate. The display pixels may be organic light-emitting diode display pixels or display pixels in a liquid crystal display. In an organic light-emitting diode display, hybrid thin-film transistor structures may be formed that include semiconducting oxide thin-film transistors, silicon thin-film transistors, and capacitor structures. The capacitor structures may overlap the semiconducting oxide thin-film transistors. Organic light-emitting diode display pixels may have combinations of oxide and silicon transistors. In a liquid crystal display, display driver circuitry may include silicon thin-film transistor circuitry and display pixels may be based on oxide thin-film transistors. A single layer or two different layers of gate metal may be used in forming silicon transistor gates and oxide transistor gates. A silicon transistor may have a gate that overlaps a floating gate structure.

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Background/Summary

(1) This application is a continuation of patent application Ser. No. 18/155,828, filed Jan. 18, 2023, which is hereby incorporated by reference herein in its entirety and which is a continuation of patent application Ser. No. 17/502,909, filed Oct. 15, 2021, now U.S. Pat. No. 11,587,954, which is hereby incorporated by reference herein in its entirety and which is a continuation of patent application Ser. No. 17/224,305, filed Apr. 7, 2021, now U.S. Pat. No. 11,177,291, which is hereby incorporated by reference herein in its entirety and which is a continuation of patent application Ser. No. 16,917,601, filed Jun. 30, 2020, now U.S. Pat. No. 10,998,344, which is hereby incorporated by reference herein in its entirety and which is a continuation of patent application Ser. No. 16/678,599, filed Nov. 8, 2019, now U.S. Pat. No. 10,741,588, which is hereby incorporated by reference herein in its entirety and which is continuation of patent application Ser. No. 16/125,973, filed Sep. 10, 2018, now U.S. Pat. No. 10,707,237, which is hereby incorporated by reference herein in its entirety and which is continuation of patent application Ser. No. 15/727,475, filed Oct. 6, 2017, now U.S. Pat. No. 10,096,622, which is hereby incorporated by reference herein in its entirety and which is a continuation of patent application Ser. No. 14/249,716, filed Apr. 10, 2014, now U.S. Pat. No. 9,818,765, which is hereby incorporated by reference herein in its entirety and which is a continuation-in-part of patent application Ser. No. 14/228,070, filed Mar. 27, 2014, now U.S. Pat. No. 9,564,478, which is hereby incorporated by reference herein in its entirety and which claims the benefit of U.S. provisional patent application Ser. No. 61/869,937, filed Aug. 26, 2013, which is hereby incorporated by reference herein in its entirety.

BACKGROUND

- (1) This relates generally to electronic devices and, more particularly, to electronic devices with displays that have thin-film transistors.
- (2) Electronic devices often include displays. For example, cellular telephones and portable computers include displays for presenting information to users.
- (3) Displays such as liquid crystal displays are formed from multiple layers. A liquid crystal display may, for example, have upper and lower polarizer layers, a color filter layer that contains an array of color filter elements, a thin-film transistor layer that includes thin-film transistors and display pixel electrodes, and a layer of liquid crystal material interposed between the color filter layer and the thin-film transistor layer. Each display pixel typically includes a thin-film transistor for controlling application of a signal to display pixel electrode structures in the display pixel.
- (4) Displays such as organic light-emitting diode displays have an array of display pixels based on light-emitting diodes. In this type of display, each display pixel includes a light-emitting diode and thin-film transistors for controlling application of a signal to the light-emitting diode.
- (5) Thin-film display driver circuitry is often included in displays. For example, gate driver circuitry and demultiplexer circuitry on a display may be formed from thin-film transistors.
- (6) If care is not taken, thin-film transistor circuitry in the display pixels and display driver circuitry of a display may exhibit non-uniformity, excessive leakage currents, insufficient drive strengths, poor area efficiency, hysteresis, and other issues. It would therefore be desirable to be able to provide improved electronic device displays.

SUMMARY

- (7) An electronic device may be provided with a display. The display may have an array of display pixels on a substrate. The display pixels may be organic light-emitting diode display pixels or display pixels in a liquid crystal display.

- (8) In an organic light-emitting diode display, hybrid thin-film transistor structures may be formed that include semiconducting oxide thin-film transistors, silicon thin-film transistors, and capacitor structures. The capacitor structures may overlap the semiconducting oxide thin-film transistors. Capacitor structures may also be formed from multiple overlapping electrode layers formed from source-drain metal layers, a polysilicon layer, and a gate metal layer may be used.
- (9) Organic light-emitting diode display pixels may have combinations of oxide and silicon transistors. Transistors such as drive transistors that are coupled to light-emitting diodes may be formed from oxide transistor structures and switching transistors may be formed from silicon transistor structures.
- (10) In a liquid crystal display, display driver circuitry may include silicon thin-film transistor circuitry and display pixels may be based on oxide thin-film transistors. A single layer or two different layers of gate metal may be used in forming silicon transistor gates and oxide transistor gates. A silicon transistor may have a gate that overlaps a floating gate structure. Oxide transistors may be incorporated into display driver circuitry.
- (11) Display driver circuitry may be configured to expose silicon transistor circuitry to lower voltage swings than oxide transistor circuitry in an array of display pixels.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a diagram of an illustrative display such as an organic light-emitting diode display having an array of organic light-emitting diode display pixels or a liquid crystal display having an array of display pixels in accordance with an embodiment.
- (2) FIG. 2 is a diagram of an illustrative organic light-emitting diode display pixel of the type that may be used in an organic light-emitting diode with semiconducting oxide thin-film transistors and silicon thin-film transistors in accordance with an embodiment.
- (3) FIG. 3 is a cross-sectional side view of illustrative thin-film transistor structures in accordance with an embodiment.
- (4) FIG. 4 is a side view of additional illustrative thin-film transistor structures in accordance with an embodiment.
- (5) FIG. 5 is a diagram of an illustrative organic light-emitting diode display pixel of the type that may include oxide and silicon thin-film transistors in accordance with an embodiment.
- (6) FIGS. 6, 7, and 8 are cross-sectional side views of illustrative thin-film transistor circuitry in a liquid crystal display in accordance with an embodiment.
- (7) FIG. 9 is a diagram of an illustrative complementary metal-oxide-semiconductor transistor inverter of the type that may be formed from a hybrid silicon-oxide transistor structure in accordance with an embodiment.
- (8) FIG. 10 is a cross-sectional side view of an illustrative thin-film transistor structure of the type that may be used to form a hybrid complementary metal-oxide-semiconductor transistor inverter in accordance with an embodiment.
- (9) FIG. 11 is a circuit diagram of gate driver circuitry in thin-film display driver circuitry in accordance with an embodiment.
- (10) FIG. 12 is a diagram of a level shifter of the type that may be used in the gate driver circuitry of FIG. 11 within display driver circuitry on a display in accordance with an embodiment.
- (11) FIG. 13 is a circuit diagram of an illustrative circuit that may be used to prevent transistors within display driver circuitry on a display from experiencing excessive voltages in accordance with an embodiment.
- (12) FIG. 14 is a cross-sectional side view of illustrative thin-film transistor circuitry in a liquid crystal display in accordance with an embodiment.

(13) FIG. **15** is a cross-sectional side view of illustrative thin-film transistor circuitry that includes a top gate semiconducting oxide transistor in a liquid crystal display in accordance with an embodiment.

(14) FIG. **16** is a cross-sectional side view of illustrative thin-film transistor circuitry that includes a top gate semiconducting oxide transistor with a light shield in a liquid crystal display in accordance with an embodiment.

(15) FIG. **17** is a cross-sectional side view of illustrative thin-film transistor circuitry that includes a top gate semiconducting oxide transistor in a liquid crystal display in accordance with an embodiment.

(16) FIG. **18** is a cross-sectional side view of illustrative thin-film transistor circuitry that includes a top gate semiconducting oxide transistor in an organic light-emitting diode display in accordance with an embodiment.

DETAILED DESCRIPTION

(17) A display in an electronic device may be provided with driver circuitry for displaying images on an array of display pixels. An illustrative display is shown in FIG. **1**. As shown in FIG. **1**, display **14** may have one or more layers such as substrate **24**. Layers such as substrate **24** may be formed from planar rectangular layers of material such as planar glass layers. Display **14** may have an array of display pixels **22** for displaying images for a user. The array of display pixels **22** may be formed from rows and columns of display pixel structures on substrate **24**. There may be any suitable number of rows and columns in the array of display pixels **22** (e.g., ten or more, one hundred or more, or one thousand or more).

(18) Display driver circuitry such as display driver integrated circuit **16** may be coupled to conductive paths such as metal traces on substrate **24** using solder or conductive adhesive. Display driver integrated circuit **16** (sometimes referred to as a timing controller chip) may contain communications circuitry for communicating with system control circuitry over path **25**. Path **25** may be formed from traces on a flexible printed circuit or other cable. The control circuitry may be located on a main logic board in an electronic device such as a cellular telephone, computer, set-top box, media player, portable electronic device, or other electronic equipment in which display **14** is being used. During operation, the control circuitry may supply display driver integrated circuit **16** with information on images to be displayed on display **14**. To display the images on display pixels **22**, display driver integrated circuit **16** may supply corresponding image data to data lines **D** while issuing clock signals and other control signals to supporting thin-film transistor display driver circuitry such as gate driver circuitry **18** and demultiplexing circuitry **20**.

(19) Gate driver circuitry **18** may be formed on substrate **24** (e.g., on the left and right edges of display **14**, on only a single edge of display **14**, or elsewhere in display **14**). Demultiplexer circuitry **20** may be used to demultiplex data signals from display driver integrated circuit **16** onto a plurality of corresponding data lines **D**. With this illustrative arrangement of FIG. **1**, data lines **D** run vertically through display **14**. Each data line **D** is associated with a respective column of display pixels **22**. Gate lines **G** run horizontally through display **14**. Each gate line **G** is associated with a respective row of display pixels **22**. Gate driver circuitry **18** may be located on the left side of display **14**, on the right side of display **14**, or on both the right and left sides of display **14**, as shown in FIG. **1**.

(20) Gate driver circuitry **18** may assert gate signals (sometimes referred to as scan signals) on the gate lines **G** in display **14**. For example, gate driver circuitry **18** may receive clock signals and other control signals from display driver integrated circuit **16** and may, in response to the received signals, assert a gate signal on gate lines **G** in sequence, starting with the gate line signal **G** in the first row of display pixels **22**. As each gate line is asserted, the corresponding display pixels in the row in which the gate line is asserted will display the display data appearing on the data lines **D**.

(21) Display driver circuitry such as demultiplexer circuitry **20** and gate line driver circuitry **18** may be formed from thin-film transistors on substrate **24**. Thin-film transistors may also be used in

forming circuitry in display pixels **22**. To enhance display performance, thin-film transistor structures in display **14** may be used that satisfy desired criteria such as leakage current, switching speed, drive strength, uniformity, etc. The thin-film transistors in display **14** may, in general, be formed using any suitable type of thin-film transistor technology (e.g., silicon-based, semiconducting-oxide-based, etc.).

(22) With one suitable arrangement, which is sometimes described herein as an example, the channel region (active region) in some thin-film transistors on display **14** is formed from silicon (e.g., silicon such as polysilicon deposited using a low temperature process, sometimes referred to as LTPS or low-temperature polysilicon) and the channel region in other thin-film transistors on display **14** is formed from a semiconducting oxide material (e.g., amorphous indium gallium zinc oxide, sometimes referred to as IGZO). If desired, other types of semiconductors may be used in forming the thin-film transistors such as amorphous silicon, semiconducting oxides other than IGZO, etc. In a hybrid display configuration of this type, silicon transistors (e.g., LTPS transistors) may be used where attributes such as switching speed and good drive current are desired (e.g., for gate drivers in liquid crystal diode displays or in portions of an organic light-emitting diode display pixel where switching speed is a consideration), whereas oxide transistors (e.g., IGZO transistors) may be used where low leakage current is desired (e.g., in liquid crystal diode display pixels and display driver circuitry) or where high pixel-to-pixel uniformity is desired (e.g., in an array of organic light-emitting diode display pixels). Other considerations may also be taken into account (e.g., considerations related to power consumption, real estate consumption, hysteresis, etc.).

(23) Oxide transistors such as IGZO thin-film transistors are generally n-channel devices (i.e., NMOS transistors). Silicon transistors can be fabricated using p-channel or n-channel designs (i.e., LTPS devices may be either PMOS or NMOS). Combinations of these thin-film transistor structures can provide optimum performance.

(24) In an organic light-emitting diode display, each display pixel contains a respective organic light-emitting diode. A schematic diagram of an illustrative organic light-emitting diode display pixel **22-1** is shown in FIG. **2**. As shown in FIG. **2**, display pixel **22-1** may include light-emitting diode **26**. A positive power supply voltage ELVDD may be supplied to positive power supply terminal **34** and a ground power supply voltage ELVSS may be supplied to ground power supply terminal **36**. The state of drive transistor **28** controls the amount of current flowing through diode **26** and therefore the amount of emitted light **40** from display pixel **22-1**.

(25) To ensure that transistor **28** is held in a desired state between successive frames of data, display pixel **22-1** may include a storage capacitor such as storage capacitor Cst. The voltage on storage capacitor Cst is applied to the gate of transistor **28** at node A to control transistor **28**. Data can be loaded into storage capacitor Cst using one or more switching transistors such as switching transistor **30**. When switching transistor **30** is off, data line D is isolated from storage capacitor Cst and the gate voltage on terminal A is equal to the data value stored in storage capacitor Cst (i.e., the data value from the previous frame of display data being displayed on display **14**). When gate line G (sometimes referred to as a scan line) in the row associated with display pixel **22-1** is asserted, switching transistor **30** will be turned on and a new data signal on data line D will be loaded into storage capacitor Cst. The new signal on capacitor Cst is applied to the gate of transistor **28** at node A, thereby adjusting the state of transistor **28** and adjusting the corresponding amount of light **40** that is emitted by light-emitting diode **26**.

(26) Organic light-emitting diode display pixels such as pixel **22-1** of FIG. **2** may use thin-film transistor structures of the type shown in FIG. **3**. In this type of structure, two different types of semiconductor are used. As shown in FIG. **3**, circuitry **72** may include display pixel structures such as light-emitting diode cathode terminal **42** and light-emitting diode anode terminal **44**. Organic light-emitting diode emissive material **47** may be interposed between cathode **42** and anode **44**. Dielectric layer **46** may serve to define the layout of the display pixel and may sometimes be referred to as a pixel definition layer. Planarization layer **50** may be formed on top of thin-film

transistor structures **52**. Thin-film transistor structures **52** may be formed on buffer layer **54** on substrate **24**.

(27) Thin-film transistor structures **52** may include silicon transistor **58**. Transistor **58** may be an LTPS transistor formed using a “top gate” design and may serve as a switching transistor in an organic light-emitting diode display pixel (see, e.g., transistor **30** in pixel **22-1** of FIG. **2**). Transistor **58** may have a polysilicon channel **62** that is covered by gate insulator layer **64** (e.g., a layer of silicon oxide). Gate **66** may be formed from patterned metal (e.g., molybdenum, as an example). Gate **66** may be covered by a layer of interlayer dielectric (e.g., silicon nitride layer **68** and silicon oxide layer **70**). Source-drain contacts **74** and **76** may contact opposing sides of polysilicon layer **62** to form the silicon thin-film transistor **58**.

(28) Thin-film transistor structures **52** may also include thin-film transistor and capacitor structures **60**. Structures **60** may include a storage capacitor (i.e., storage capacitor Cst of FIG. **2**) and an oxide thin-film transistor structure. The storage capacitor may have a first terminal (sometimes referred to as a plate, electrode, or electrode layer) that is formed from polysilicon layer **62'** (patterned as part of the same layer as layer **62**). Gate insulator layer **64'**, which may be an extended portion of gate insulator layer **64**, may cover terminal **62'**. The capacitor may have a second terminal formed from metal layer **66'**. Metal layer **66'** may be patterned from the same metal layer that is used in forming gate **66** of transistor **58**. Dielectric layers **68** and **70** may cover metal layer **66'**. The thin-film transistor in structures **60** may be a “bottom gate” oxide transistor. Layer **66'**, which serves as the second terminal of capacitor Cst (i.e., node A of FIG. **2**) may also serve as the gate of the oxide transistor. The oxide transistor may serve as drive transistor **28** of FIG. **2**. The “gate insulator” of the oxide transistor may be formed from the layer of interlayer dielectric (i.e., layers **68** and **70**). The channel semiconductor of the oxide transistor may be formed from oxide layer **80** (e.g., IGZO). Oxide layer **80** may overlap polysilicon capacitor electrode layer **62'** (i.e., the oxide transistor may overlap the capacitor), thereby saving space. Source-drain terminals **82** and **84** may be formed from metal contacting opposing ends of semiconducting oxide layer **80**.

(29) Transistors such as LTPS transistors and oxide transistors may be formed with different layouts. For example, LTPS transistors tend to have high carrier mobilities. As a result, LTPS transistors may have relatively long gate lengths L and relatively short gate widths to ensure appropriately low ratios of W/L to compensate for the relatively high mobility of these transistors. This may cause LTPS transistors to be relatively inefficient for pixel layout. Oxide transistors may be constructed with W/L ratios with smaller aspect ratios (e.g., 4/4 for oxide relative to 3/30 for LTPS). Due to these layout efficiency considerations, it may be preferred to use oxide transistors as the drive transistors in display pixels **22-1**. The relatively fast switching speed provided by LTPS transistor may make it preferable to use LTPS transistors for switching transistors such as transistor **30** of FIG. **2**.

(30) In display pixels with more transistors (e.g., three or more, four or more, five or more, six or more, seven or more, or eight or more), the selection of which transistors are implemented using LTPS technology and which transistors are implemented using oxide technology may be made so as to balance transistor performance considerations between the two types of transistors.

(31) When implementing driving transistors, LTPS transistors tend to exhibit larger size (longer channel length) than oxide transistors, tend to exhibit larger dark currents than oxide transistors, and may exhibit poorer uniformity than oxide transistors. LTPS driving transistors may also exhibit more hysteresis than oxide driving transistors. As a result of these factors, it may often be advantageous to form driving transistors in an organic light-emitting diode display pixel from oxide transistors. The oxide driving transistors may exhibit low leakage current and minimal hysteresis.

(32) When implementing switching transistors, LTPS transistors may be smaller than oxide transistors, may exhibit smaller amounts of parasitic capacitance than oxide transistors, and may exhibit lower power consumption than oxide transistors. As a result of factors such as these, it may

often be advantageous to form switching transistors in an organic light-emitting diode display pixel from LTPS transistors. The LTPS switching transistors may exhibit high switching speed and low parasitic capacitance.

(33) An illustrative hybrid thin-film transistor structure that may be used in implementing both LTPS and oxide transistors in a single organic light-emitting diode display pixel (e.g., to implement a circuit such as display pixel circuit **22-1** of FIG. **2**) is shown in FIG. **4**. Hybrid thin-film transistor structures **114** of FIG. **4** include silicon thin-film transistor **108**, capacitor (Cst) **110**, and oxide transistor **112**. Silicon transistor **108** is formed from polysilicon layer **90**. Gate insulator layer **92** covers polysilicon layer **90**. A layer of gate metal is patterned on top of gate insulator layer **92** to form gate **94**, capacitor electrode **96**, and gate electrode **98**. A layer of interlayer dielectric material such as silicon nitride layer **116** and silicon oxide layer **118** may cover the patterned gate metal structures. Source-drain contacts **100** and **94** for silicon transistor **108** may contact (i.e., may be shorted to) polysilicon layer **90** in the vicinity of channel region **106**. Gate **94** of transistor **108** may serve as an implant mask to allow low-density drain implants to be formed in polysilicon layer **90** in regions **104** adjacent to polysilicon channel region **106** of transistor **108**.

(34) Source-drains **100** and **102** of silicon transistor **108**, capacitor electrode **120**, and source-drains **122** and **124** of oxide transistor **112** may be formed from patterned portions of a common metal layer on interlayer dielectric **116** and **118**.

(35) Capacitor **110** may have a first terminal formed from metal electrode **120** and from portion **126** of polysilicon layer **90**. Capacitor **110** may have a second terminal formed from metal electrode **96**.

(36) Oxide transistor **112** may have a semiconductor oxide layer such as an IGZO layer **128**, source-drain contacts **122** and **124**, and gate **98**. Gate **98** is separated from semiconductor oxide **128**, which serves as the channel region for transistor **112** by dielectric **116** and **118**. Dielectric **116** and **118** therefore serves as the gate insulator for oxide transistor **112**.

(37) FIG. **5** is a circuit diagram of another illustrative organic light-emitting diode pixel circuit that may be used in display **14**. Pixel **22-2** includes driver transistor **28** for driving current into light-emitting diode **26**. Storage capacitor Cst is used to store signals on the gate of transistor **28** between frames. Sensing line SENSING used to implement a compensation scheme to adjust for pixel-to-pixel variations in transistor performance. Gate lines SCAN and SCAN2 are used in applying control signals to switching transistors **30-1** and **30-2**.

(38) To optimize performance in display pixel **22-2**, it may be desirable to use hybrid structures of the type shown in FIGS. **3** and **4** or other configurations for forming silicon and/or oxide thin-film transistors and capacitors. For example, it may be desirable to form drive transistor **28** from an oxide transistor (e.g., an NMOS oxide transistor), while forming switching transistors such as transistors **30-1** and **30-2** from silicon transistors or from a mixture of silicon (NMOS and/or PMOS) and oxide (NMOS) transistors.

(39) With a first illustrative configuration, transistor **30-1** is an oxide transistor, transistor **30-2** is an oxide transistor, and transistor **28** is an oxide transistor. With a second illustrative configuration, transistor **30-1** is a silicon transistor, transistor **30-2** is a silicon transistor, and transistor **28** is an oxide transistor. A hybrid transistor structure such as the structure of FIG. **3** or the structure of FIG. **4** may be used in this scenario (e.g., to implement transistors **30-1** and **28** and capacitor Cst). With an illustrative third configuration, transistor **30-1** is a silicon transistor, transistor **30-2** is an oxide transistor, and transistor **28** is an oxide transistor. As with the second illustrative configuration, a hybrid transistor structure such as the structure of FIG. **3** or the structure of FIG. **4** may be used to implement transistors **30-1** and **28** and capacitor Cst.

(40) If desired, display **14** may be a liquid crystal display. In this type of scenario, each pixel of display **14** may contain an electrode structure for applying an electric field to an associated portion of a liquid crystal layer in the display, a capacitor for storing charge on the electrode between frames of image data, and a thin-film transistor for controlling the application of the electric field to

the electrodes. With one suitable arrangement, gate driver circuitry **18** and demultiplexer circuitry **20** (FIG. **1**) in the liquid crystal display may be formed from silicon transistors and the thin-film transistors in display pixels **22** may be formed from oxide transistors. The silicon transistors have high mobility channel regions and are well suited for fast switching speeds and high drive currents while operating at low voltages and low power. The oxide thin-film transistors in display pixels **22** exhibit low leakage currents.

(41) Thin-film transistor structures of the type that may be used in forming a liquid crystal display with both silicon and oxide transistors are shown in FIG. **6**. As shown in FIG. **6**, thin-film transistor structures **242** may include silicon thin-film transistor structures **216** (e.g., for forming parts of peripheral circuits such as display driver circuitry **18** and demultiplexer circuitry **20**) and oxide thin-film transistor structures **240** (e.g., for forming display pixels **22** in a liquid crystal display having a layout of the type shown by display **14** of FIG. **1**).

(42) Structures **216** and **240** may be formed on buffer layer **202** on substrate **24**. Polysilicon layer **204** may be deposited on buffer **202**. Gate insulator layer **206** may be formed on polysilicon layer **204**. A common layer of metal may be patterned to form metal structures **218**, **220**, and **228**.

Structure **218** may serve as the gate for a silicon transistor that includes source-drain contacts **212** and **214** and a channel formed from polysilicon **204**. Metal structure **228** may serve as a gate for an oxide transistor formed from semiconducting oxide layer **224** (e.g., IGZO) and source-drain terminals **222** and **226**. Metal structure **228** may also serve as a light shield that helps block backlight in display **14** from reaching oxide layer **224**, so no separate light shielding structures need be incorporated in structures **240**. Interlayer dielectric such as silicon nitride layer **208** and **210** may cover gate **218** in structure **216** and may serve as a gate insulator for gate **228** in the oxide transistor of structures **240**.

(43) Metal **230** contacts source-drain **226** of the display pixel thin-film oxide transistor that is formed from oxide layer **224**. Metal **230** may be supported by organic layer **232**. On the surface of organic layer **232**, metal **230** may form an electrode with multiple fingers. Dielectric layer **236** may isolate electrode **230** from common electrode (Vcom) **234**. During operation, electric fields are produced between electrode **230** and electrode **234**. These fields pass through the liquid crystal material in the display. If desired, display **14** may incorporate capacitive touch sensors that are formed from portions of Vcom electrode **234**. In this type of configuration, optional metal lines such as line **238** may be used to help reduce the resistance of the material used in forming electrode **234** (which may be, for example, a somewhat resistive conducting material such as indium tin oxide).

(44) The thickness of layers **208** and **210** may be about 6000 angstroms. This relatively large thickness may help minimize capacitance between gate **218** and nearby metal structures such as source-drain **214**, but may limit switching speeds in the oxide transistor. To address this concern, a design of the type used by structures **242'** in FIG. **7** may be used. With the FIG. **7** arrangement, an additional semiconductor fabrication mask may be used to create a gate for the oxide transistor that is formed from a separate layer of metal from the metal layer used in forming gate **218**. With this approach, only a single 3000 angstrom dielectric layer **210'** (formed, e.g., from sublayers of silicon nitride and silicon oxide) is used to separate oxide transistor gate **228'** from oxide layer **224**, so oxide transistor switching speed may be enhanced. The arrangement of structures **242'** of FIG. **7** allows gate **218** and gate **228'** to be formed from different metals. For example, gate **218** may be formed from a refractory metal such as Mo to accommodate the elevated temperatures associated with activating the silicon transistor, whereas gate **228'** may be formed from a lower resistance metal such as copper.

(45) In some applications, the handling of high drive voltages (gate-to-source and drain) may need to be considered. Transistor structures **242''** of FIG. **8** may be used in scenarios in which it is desired to handle relatively large (e.g., 20 volt) swings on the silicon transistor gate. In this situation, gate insulator layer **206** may be insufficiently thin to withstand damage from a 20 volt

signal. For example, gate insulator **206** may be about 800 angstroms thick, which may not be sufficiently thick to reliably handle 20 volt drive voltages. To ensure that gate insulator layer **206** is not overly stressed, gate structure **218** may be converted into a floating (electrically isolated) metal structure and an additional metal layer (i.e., part of the same metal layer that is patterned to form gate **228'** of oxide transistor **240**) may be used in forming silicon transistor gate **218'**. Floating gate **218** may be retained to serve as a mask for low density drain (LDD) implants made into the source and drain contact portions of polysilicon layer **204**, even though floating gate **218** is not driven with control signals during operation of silicon transistor **216**.

(46) In a hybrid silicon/oxide liquid crystal display, it is not necessary to form display driver circuitry such as gate driver circuitry **18** and demultiplexer circuitry **20** from silicon transistors. If desired, some of this display driver circuitry may be formed from oxide transistors. For example, low drive current CMOS-type circuits in the peripheral circuitry of display **14** such as illustrative CMOS inverter **300** of FIG. **9** may include oxide transistors. It may be challenging to form PMOS oxide transistors, so circuits such as inverter **300** may, if desired, be formed using an NMOS oxide transistor and a PMOS silicon transistor (as an example).

(47) Hybrid oxide-silicon thin-film transistor structures such as illustrative thin-film transistor structures **302** of FIG. **10** may be used in forming CMOS-type circuitry in display driver circuitry such as gate driver circuitry **18** and demultiplexer circuitry **20**. As shown in FIG. **10**, structures **302** may have a polysilicon layer **308** that is formed on substrate **24**. P-channel active area **310** may be formed under gate **312**. Gate insulator layer **306** (e.g., silicon oxide) may separate gate **312** from silicon channel region **310** in silicon layer **308**. Dielectric layer **302** (e.g., sublayers of silicon oxide and silicon nitride) may cover gate **312**. Dielectric layer **306** may separate gate **312** from overlapping oxide layer **312**. Oxide layer **312** may be a semiconducting oxide such as IGZO material. Gate **312** may be formed from a first patterned metal layer. A second patterned metal layer may be used in forming output terminal **322**, source terminal **316**, and drain terminal **318**. Passivation layer **320** may cover terminals **316** and **312**. Gate **312** may be formed from materials such as molybdenum, molybdenum tungsten, tungsten, or other metals. Metal for forming structures such as metal structures **322**, **316**, and **318** may be formed from metal such as aluminum, molybdenum, etc.

(48) With the arrangement of FIG. **10**, gate **314** serves as a common (shared) gate for two transistors. In particular, gate **314** (see, e.g., terminal Vin of FIG. **9**) serves as both a gate for a PMOS silicon transistor (transistor TP of FIG. **9**) that is formed from silicon layer **308** and as a gate for an NMOS oxide transistor (transistor TN of FIG. **9**) that is formed from oxide layer **312**. Oxide layer **312** is located above gate **314** and silicon layer **310** is located below gate **314**. The shared gate arrangement of FIG. **10** allows a CMOS inverter of the type shown in FIG. **9** to be implemented compactly.

(49) FIG. **11** shows illustrative gate driver circuitry **18** that may be used on a liquid crystal display. Circuitry **18** may use signals with a relatively small voltage swing (e.g., a 15 volt or 16 volt swing) for silicon transistors while producing gate signals G with a larger voltage swing (e.g., a 20 volt swing or more) to ensure satisfactory operation of oxide thin-film transistors in display pixels **22** that are being driven by the gate signals.

(50) As shown in FIG. **11**, circuitry **18** may have a shift register formed from a series of linked SR latches **400** or other register circuits. Each row of circuitry in FIG. **11** is associated with a separate row of display pixels **22** in a liquid crystal display and provides a respective gate signal G for that row of display pixels. During operation, trigger signal TRIGGER may be applied to the latch in the first row of the shift register in circuitry **18** while a clock signal LOAD CLOCK is being applied to the shift register. The trigger signal causes a cascading signal to ripple down through the shift register. In response, each latch **400** asserts its output OUT in sequence. Each row of gate driver circuitry **18** has a respective level shifter **402** and buffer **404** that receive output signal OUT.

(51) Output signal OUT ranges from a high voltage of 15 V (or other suitable voltage) to 0 volts (or

other suitable voltage). The 15 volt swing that is associated with this type of configuration can be tolerated by silicon thin-film transistors in latches **400**, whereas larger voltage swings such as 20 volt swings might overly stress the silicon thin-film transistors. Level shifter **402** shifts the 15 volt to 0 volt signal OUT from latch **400** so that the output on path **406** from level shifter **402** ranges from 5 volts to -11 volts (i.e., a swing of 16 volts that can be tolerated by the silicon transistors in level shifter **402**). Buffer **404** receives the 15 volt to 0 volt signal OUT from latch **400** as input signal IN_H and receives the 5 volt to -11 volt signal as input signal IN_L. Buffer **404** preferably contains silicon thin-film transistors. The design of buffer **404** allows buffer **404** to produce an output signal (gate line signal G) with a large voltage swing (e.g., 15 volts to -11 volts) of the type that is appropriate for controlling oxide transistors in the array of display pixels **22** on the liquid crystal display.

(52) FIG. **12** is a circuit diagram of an illustrative circuit of the type that may be used to implement level shifter **402**. Signals from output OUT of latch **400** may be received at input **410** of level shifter **402** and corresponding level-shifted output signals (signals IN_L) for buffer **404** may be provided at output **412** of level shifter **402**. Other level shifter designs may be used for level shifter **402** if desired. The configuration of FIG. **12** is merely illustrative. Silicon thin-film transistors may be used in forming level shifter **402**.

(53) Circuitry **414** of FIG. **13** is an example of a design that may be used in implementing buffer **404** of FIG. **11**. With this design, signals IN_H and IN_L are identical square wave pulses with different respective voltage swings. Signal IN_H ranges from 15 to 0 volts. Signal IN_L ranges from 5 to -11 volts. Corresponding output signal (gate line signal) G in this example is a square wave pulse that ranges from 15 volts to -11 volts and therefore has a swing of more than 20 volts.

(54) Ground voltage GND is applied to the gates of transistors T2 and T3. This limits that maximum voltage experienced by the transistors of circuit **414** to less than about 16 volts, even though the output swing of circuit **414** is more than 20 volts. The ground voltage GND on the gates of transistors T2 and T3 causes these transistors to turn off to protect transistors T1 and T4 whenever excessive source terminal voltage swing is detected. Consider, as an example, transistors T1 and T2. Transistor T2 may be characterized by a threshold voltage V_{th} . If the source S of transistor T1 starts to fall below voltage $GND - V_{th}$, transistor T2 will turn off and isolate transistor T1. Transistors T3 and T4 operate in the same way. Using this arrangement, none of the transistors in buffer **414** is exposed to excessive voltage swings, allowing transistors T1, T2, T3, and T4 to be formed from silicon thin-film transistors.

(55) If desired, other circuit configurations may be used to allow gate driver circuitry **18** to operate in an environment in which gate line signal G has a large voltage swing to accommodate oxide transistors in display pixels **22**. As an example, a subset of the level shifter transistors and a subset of the output buffer transistors may be implemented using oxide thin-film transistor structures in addition to using silicon thin-film transistor structures.

(56) FIG. **14** is a cross-sectional side view of additional thin-film transistor circuitry of the type that may be used in a liquid crystal display. As shown in FIG. **14**, thin-film transistor structures **242** may include silicon thin-film transistor structures **216** (e.g., for forming parts of peripheral circuits such as display driver circuitry **18** and demultiplexer circuitry **20**) and oxide thin-film transistor structures **240** (e.g., for forming display pixels **22** in a liquid crystal display having a layout of the type shown by display **14** of FIG. **1**).

(57) Structures **216** and **240** may be formed on buffer layer **202** on substrate **24**. Polysilicon layer **204** may be deposited on buffer **202**. Gate insulator layer **206** may be formed on polysilicon layer **204**. A common layer of metal may be patterned to form metal structures **218**, **220**, and **228**. Structure **218** may serve as the gate for a silicon transistor that includes source-drain contacts **212** and **214** and a channel formed from polysilicon **204**. Metal structure **228** may serve as a gate for an oxide transistor formed from semiconducting oxide layer **224** (e.g., IGZO) and source-drain terminals **222** and **226**. Metal structure **228** may also serve as a light shield that helps block

backlight in display **14** from reaching oxide layer **224**, so no separate light shielding structures need be incorporated in structures **240**. Interlayer dielectric such as silicon nitride layer **208** and **210** may cover gate **218** in structure **216** and may serve as a gate insulator for gate **228** in the oxide transistor of structures **240**.

(58) Metal structures **218**, **220**, and **228** and routing lines such as interconnect line **502** may be formed from a first metal layer (sometimes referred to as an M1 layer). Metal **222** and **226**, which form source-drain contacts for the oxide transistor of structures **240**, and routing lines such as interconnect line **500** may be formed from a second metal layer (sometimes referred to as an SD1 layer). Metal structures **212**, **214**, and routing lines such as interconnect line **506** may be formed from a third metal layer (sometimes referred to as an SD2 layer). Dielectric layers **232B** may separate the second metal layer from the third metal layer. Dielectric layer **232A** may separate the third metal layer from metal structures such as metal layer **234**.

(59) Metal **230** contacts metal layer **504** and is thereby coupled to source-drain **226** of the display pixel thin-film oxide transistor that is formed from oxide layer **224**. Metal **230** may be supported by organic layer **232B**. On the surface of organic layer **232B**, metal **230** may form an electrode with multiple fingers. Dielectric layer **236** may isolate electrode **230** from common electrode (Vcom) **234**. During operation, electric fields are produced between electrode **230** and electrode **234**. These fields pass through the liquid crystal material in the display. If desired, display **14** may incorporate capacitive touch sensors that are formed from portions of Vcom electrode **234**. In this type of configuration, optional metal lines such as line **238** may be used to help reduce the resistance of the material used in forming electrode **234** (which may be, for example, a somewhat resistive conducting material such as indium tin oxide).

(60) Capacitive coupling between the routing lines in display **14** can lead to switching losses. As an example, source-drain structure **222** may be coupled to the data line in display **14**. The voltage on this line switches relative to Vcom (electrode **234**) and can lead to power losses. The presence of dielectric layers **232A** and **232B** can help reduce capacitive coupling between the data line and Vcom electrode and thereby reduce power losses. The presence of these dielectric layers can also reduce capacitive coupling between routing lines in display **14** (e.g., capacitive coupling between routing lines and other structures of the first and second metal layers, the first and third metal layers, etc.). Layers **232A** and **232B** may be formed from low-dielectric-constant organic dielectric or other dielectric material. As an example, layers **232A** and **232B** may be acrylic polymers, other polymers, dielectrics of the type sometimes referred to as spin-on-glass, (e.g., spin-on-glass polymers deposited via slit coating tools, etc.), siloxane-based materials, etc.

(61) FIG. **15** is a cross-sectional side view of illustrative thin-film transistor circuitry for a liquid crystal display that includes a top gate semiconducting oxide transistor. As shown in FIG. **15**, thin-film transistor structures **242** may include silicon thin-film transistor structures **216** and semiconducting oxide thin-film transistor structures **240**. Silicon thin-film transistor structures **216** may be used in peripheral circuits such as display driver circuitry **18** and demultiplexer circuitry **20** and/or may be used in forming circuits for display pixels **22** in a liquid crystal display.

Semiconducting oxide thin-film transistor structures **240** may be used in peripheral circuits such as display driver circuitry **18** and demultiplexer circuitry **20** and/or may be used in forming circuits for display pixels **22** in a liquid crystal display. Transistors such as silicon (polysilicon) transistor **216** may be n-channel or p-channel devices. Transistors such as semiconducting oxide transistor **240** may be n-channel or p-channel devices.

(62) Structures **216** and **240** may be formed on buffer layer **202** on substrate **24**. Buffer layer **202** may be formed from a dielectric such as an inorganic dielectric. Buffer layer **202** may help prevent ions in substrate **24** from migrating into structures **216** and **240**.

(63) Polysilicon layer **204** may be deposited on buffer **202**. Gate insulator layer **206** may be formed on polysilicon layer **204**. Gate insulator layer **206** may be formed from a dielectric such as silicon oxide (e.g., a 100 nm silicon oxide layer). A common layer of metal may be patterned to form metal

structures **218**, **220**, and **228**. Structure **218** may serve as the gate for a silicon transistor that includes source-drain contacts **212** and **214** and a channel formed from polysilicon **204**. Metal structure **228** may serve as a gate for a top gate oxide transistor (i.e., a semiconducting oxide transistor) formed from semiconducting oxide layer **224** (e.g., IGZO) and source-drain terminals **222** and **226**. One or more layers of interlayer dielectric (ILD) may cover metal structures **218**, **220**, and **228**. For example, a first dielectric layer such as layer **208** and a second dielectric layer such as layer **210** may cover metal structures **218**, **220**, and **228**. Layer **208** may be a silicon nitride layer and layer **210** may be a silicon oxide layer (as examples). Because there is no lateral overlap between gate **228** and source-drain electrodes **222** and **226**, parasitic capacitance between gate **228** and source-drain structures **222** and **226** may be minimized. Moreover, layers **208** and **210** of the oxide transistor of FIG. **15** may be thicker than layers **208** and **210** in bottom-gate oxide transistor of FIG. **14**, thereby further reducing parasitic capacitances.

(64) Metal structures **218**, **220**, and **228** may be formed from a first metal layer (sometimes referred to as an M1 layer). Metal **222** and **226**, which form source-drain contacts for the oxide transistor of structures **240** and metal **212** and **214**, which form source-drain contacts for the silicon transistor of structures **216** may be formed from a second metal layer (sometimes referred to as an SD1 layer or M2 layer). Metal structures such as metal line **238** may be formed from a third metal layer (sometimes referred to as an M3 layer). Dielectric **232** (e.g., an organic dielectric layer such as a polymer layer) may separate the second metal layer from the third metal layer.

(65) Metal **230** contacts source-drain **226** of the display pixel thin-film oxide transistor that is formed from oxide layer **224**. Metal **230** may be supported by organic layer **232**. On the surface of organic layer **232**, metal **230** may form an electrode with multiple fingers (e.g., a pixel electrode for a display pixel in the display). Dielectric layer **236** may isolate electrode **230** from common electrode (Vcom) **234**. During operation, electric fields are produced between electrode **230** and electrode **234**. These fields pass through the liquid crystal material in the display that is formed on top of the structures of FIG. **15**. If desired, display **14** may incorporate capacitive touch sensors that are formed from portions of Vcom electrode **234**. In this type of configuration, optional metal lines such as line **238** may be used to help reduce the resistance of the material used in forming electrode **234** (which may be, for example, a somewhat resistive conducting material such as indium tin oxide).

(66) As shown in FIG. **16**, an optional light shielding structure such as light shield **520** can be formed under semiconducting-oxide transistor **240** or elsewhere in the display. Light shield **520** may be formed from an opaque material such as a metal, an oxidized metal, a dark polymer, or other light blocking materials. The presence of light shield **520** can help prevent stray light from disrupting the operation of semiconducting oxide transistor structures **240** or other overlapping structures.

(67) In the example of FIG. **17**, dielectric layer **232** of FIG. **15** has been divided into two dielectric layers **232A** and **232B**. Layer **232A** may overlap the source-drain electrodes of transistors **216** and **240**. Layer **232B** may be interposed between the source-drain electrodes and other metal structures formed from the source-drain metal layer and layers **208** and **210**. As described in connection with FIG. **14**, this type of two-layer approach can reduce capacitive coupling between the metal structures of devices **216** and **240**. A cross-sectional side view of illustrative thin-film transistor circuitry that includes a top gate semiconducting oxide transistor in an organic light-emitting diode display is shown in FIG. **18**. As shown in FIG. **18**, circuitry **72** may include display pixel structures such as light-emitting diode cathode terminal **42** and light-emitting diode anode terminal **44**. Organic light-emitting diode emissive material **47** may be interposed between cathode **42** and anode **44**. Pixel definition layer **46** may be a dielectric layer **46** that serves to define the layout of the display pixel. Layer **46** may be formed from a polymer such as a black polymer to help block stray light.

(68) Planarization layer **50** may be formed on top of thin-film transistor structures **52**. Thin-film

transistor structures **52** may be formed on buffer layer **54** on substrate **24**. Substrate **24** may be formed from metal, glass, polymer, other materials, or combinations of these materials. Buffer layer **54** may be formed from an inorganic dielectric layer that helps prevent ions in substrate **24** from disrupting the operation of structures **52**. Optional functional layer **522** may be interposed between buffer layer **54** and substrate **24**. Functional layer **522** may be a stress relief layer, a light-blocking layer, a layer used in forming components such as capacitors (e.g., capacitor electrodes for pixel circuits and/or peripheral circuits), etc.

(69) Thin-film transistor structures **52** may include silicon transistor **58**. Transistor **58** may be an LTPS transistor formed using a top gate design and may serve as a switching transistor in an organic light-emitting diode display pixel (see, e.g., transistor **30** in pixel **22-1** of FIG. **2**). Transistor **58** may also be used in peripheral circuits (e.g., driver circuitry **18** and demultiplexer circuitry **20**).

(70) Transistor **58** may have a polysilicon channel **62** that is covered by gate insulator layer **64** (e.g., a layer of silicon oxide having a thickness of 100 nm or other suitable thickness). Gate **66** may be formed from patterned metal (e.g., molybdenum, as an example). Gate **66** may be covered by a layer of interlayer dielectric (e.g., silicon nitride layer **68** and silicon oxide layer **70**). Source-drain contacts **74** and **76** may contact opposing sides of polysilicon layer **62** to form silicon thin-film transistor **58**.

(71) Dielectric layer **526** may cover source-drain structures **74** and **76**. Optional metal layer **524** may be formed on layer **526** and may, if desired, contact underlying metal structures through vias (see, e.g., vias **528**). Structure **66** may be formed in a first (“M1”) metal layer. Source-drain electrodes **74** and **76** may be formed in a second metal layer. Metal layer **524** may be formed as part of a third (“M3”) metal layer. Layer **524** may overlap portions of transistor **58** and/or transistor **60** and may be used for forming capacitors or signal interconnect lines (i.e., routing). Layer **524** may be overlapped by emissive material layer **47** and may form a light-blocking structures that prevent stray light from emissive material **47** from reaching underlying transistor structures, etc.

(72) Thin-film transistor structures such as semiconducting-oxide thin-film transistor structures **60** and silicon thin-film transistor structures **58** may be used in forming part of a pixel circuit in an organic light-emitting diode display and/or may be used in forming part of peripheral circuitry **18** and **20**. Thin-film transistor **60** of FIG. **18** may be a top gate semiconducting-oxide transistor. The gate insulator layer **64**, which serves as the gate insulator for silicon transistor **58**, also serves as the gate insulator for oxide transistor **60**.

(73) Metal gate **532** forms the gate of oxide transistor **60**. The channel semiconductor of the oxide transistor may be formed from semiconducting oxide layer **128** (e.g., IGZO). Source-drain terminals **534** and **536** may be formed from metal contacting opposing ends of semiconducting oxide layer **128**. Metal structures **530** and **538** may be used for routing and may be formed from the same layer of metal that is patterned to form gates **66** and **532**. Structures such as source-drain structures **534** and **536** may be formed from the same layer of metal that is used in forming source-drain structures **74** and **76**.

(74) The foregoing is merely illustrative and various modifications can be made by those skilled in the art without departing from the scope and spirit of the described embodiments. The foregoing embodiments may be implemented individually or in any combination.

Claims

1. A display pixel comprising: a light-emitting diode; a plurality of transistors that is configured to control the light-emitting diode, wherein the plurality of transistors includes a semiconducting oxide transistor having a first gate and a silicon transistor having a second gate and wherein the semiconducting oxide transistor and the silicon transistor are switching transistors; a capacitor coupled to the plurality of transistors, wherein the capacitor has an electrode layer; a layer of metal

that is patterned to form the second gate and the electrode layer, wherein the semiconducting oxide transistor comprises a layer of semiconducting oxide and wherein the first gate and the layer of semiconducting oxide are overlapping; a first dielectric layer formed over the layer of metal; and a second dielectric layer formed over the first dielectric layer.

2. The display pixel defined in claim 1, wherein the first gate comprises molybdenum.
3. The display pixel defined in claim 1, wherein the first gate comprises molybdenum and a refractory metal that is different from molybdenum.
4. The display pixel defined in claim 3, wherein the refractory metal that is different from molybdenum comprises tungsten.
5. The display pixel defined in claim 1, wherein the capacitor is a storage capacitor configured to store data associated with a frame.
6. The display pixel defined in claim 1, wherein an additional layer of metal is interposed between the first and second dielectric layers.
7. A display pixel comprising: a light-emitting diode; a plurality of transistors that is configured to control the light-emitting diode, wherein the plurality of transistors includes a semiconducting oxide transistor and a silicon transistor; a capacitor coupled to the plurality of transistors; a layer of metal that forms a gate for the silicon transistor and an electrode layer for the capacitor; a first dielectric layer formed over the layer of metal; and a second dielectric layer formed over the first dielectric layer.
8. The display pixel defined in claim 7, wherein an additional layer of metal is interposed between the first and second dielectric layers.
9. The display pixel defined in claim 7, wherein the first dielectric layer is silicon oxide.
10. The display pixel defined in claim 7, wherein the first dielectric layer is silicon nitride.
11. The display pixel defined in claim 7, wherein the capacitor is a storage capacitor configured to store data associated with a frame.
12. The display pixel defined in claim 7, wherein the semiconducting oxide transistor comprises an additional gate that comprises molybdenum.
13. A display that comprises a display pixel, wherein the display pixel comprises: a light-emitting diode; a semiconducting oxide transistor having a first gate, wherein the semiconducting oxide transistor is coupled between a voltage line and a terminal of the drive transistor; a silicon transistor having a second gate; a capacitor having an electrode layer; a layer of metal that is patterned to form the second gate and the electrode layer, wherein the semiconducting oxide transistor comprises a layer of semiconducting oxide and wherein the first gate and the layer of semiconducting oxide are overlapping; a first dielectric layer formed over the layer of metal; and a second dielectric layer formed over the first dielectric layer.
14. The display defined in claim 13, wherein the first gate comprises molybdenum.
15. The display defined in claim 13, wherein the first gate comprises molybdenum and a metal that is different from molybdenum.
16. The display defined in claim 13, wherein the refractory metal that is different from molybdenum comprises tungsten.
17. The display defined in claim 13, wherein the capacitor is a storage capacitor configured to store data associated with a frame.
18. The display defined in claim 13, wherein an additional layer of metal is interposed between the first and second dielectric layers.
